Reversing Protectionism: A First Look at Product-level Trade Data from Smoot-Hawley to the GATT*

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Abstract

We provide the first comprehensive product-level data on—and a systematic analysis of—the largest reversal of U.S. tariffs in the last century. Spanning the Smoot-Hawley Tariff Act of 1930 through the first GATT round in 1947, our dataset pairs U.S. tariff schedules with product-level import records. Using ad valorem equivalents that combine specific, compound, and mixed tariffs with ad valorem rates, we show that the tariff reductions from the bilateral agreements under the Reciprocal Trade Agreements Act and the first GATT round were both broad-based and heterogeneous. This setting yields the first empirical test of the terms-of-trade theory for a large country moving from unilateral to cooperative trade policy. Products with higher initial U.S. import values received systematically larger tariff cuts, consistent with importer market power. These effects are larger than those estimated for economies acceding to the modern WTO.

JEL Codes: F13, F14, N72

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1 Introduction

An important topic in international economics is understanding what motivates countries to liberalize their trade policy, and in particular, to form trade agreements.[1] Given the recent return to broad-based import protection policies in the U.S. and elsewhere—to levels not seen since the mid-twentieth century—it is more important than ever that we understand the historical dynamics behind the U.S. trade liberalization between the previous peak in U.S. tariffs under Smoot-Hawley in 1930 and the first GATT round in 1947. This paper provides the data to make this possible. It then uses that data to characterize the trade liberalization of the U.S. as it was building the cooperative trading system that culminated in the WTO.

In particular, we provide an empirical test of the terms-of-trade (TOT) theory for the U.S.—a country that was large in terms of its importer market power—as it moved from unilateral to cooperative trade policy. The TOT theory posits that, when countries possess importer market power, they may set higher import tariffs to shift the cost of protecting domestic industries onto their trading partners. Acting on this importer market power results in unilateral tariffs that are generally higher than the level that optimizes the country's welfare. Trade agreements then lead to lower, more efficient tariff levels. While the theoretical literature has received much attention dating back to at least Johnson (1953), there is less research providing empirical evidence of the TOT theory.[2]

One reason empirical evidence is limited is that the TOT theory applies to countries moving from a non-cooperative to a cooperative trade policy setting. With the GATT/WTO spanning multiple decades and the proliferation of preferential trade agreements, observing a non-cooperative setting is particularly challenging for the case of a large economy. This is because comprehensive, product-level panel data for tariffs and imports has not been available for the time period preceding the GATT. For example, Bown and Irwin (2017) document the available data covering the GATT rounds and find it to be frustratingly

[1] See Maggi (2014) for a detailed overview of the motives for trade agreements.

[2] For the empirical literature, see Broda, Limao, and Weinstein (2008); Bagwell and Staiger (2011); Bown and Crowley (2013); Ludema and Mayda (2013); Bown and Tovar (2016); Irwin and Soderbery (2021); Jestrab (2024); and Blanchard, Bown, and Johnson (2026).

sparse.[3] Broda, Limao, and Weinstein (2008) and Bagwell and Staiger (2011) use non-WTO members to proxy for non-cooperative tariffs; however, this yields samples that are predominantly "small" and developing economies. These samples are not ideal for testing the TOT theory, since "small" economies by definition lack the importer market power that drives the theory.

Our empirical test of the terms-of-trade (TOT) theory is the first for a large country moving from unilateral to cooperative trade policy. Digitizing and concording tariff schedules and import records from archival sources, we construct a detailed account of the United States' gradual transition from a non-cooperative regime of historically high tariff protection to the cooperative framework embedded in a global trading system. Our evidence shows that U.S. tariff changes between 1930 and the first GATT round in 1947 were consistent with predictions from the TOT theory, despite distinctive institutional constraints.

We locate, digitize, and concord the U.S. tariff schedules covering the universe of potentially imported products from the 1930 Smoot-Hawley Act through the first GATT round in 1947. The resulting panel captures each product's tariff under unilateral actions, bilateral trade agreements, and the GATT. We pair this with U.S. import data for select years, building concordances that track ad valorem equivalent tariffs (AVEs), import values, and quantities at the 7-digit Schedule A product level. This panel is a contribution in its own right, providing a resource for future research on this formative period in U.S. and global trade policy.[4]

This new panel data set allows us to document stylized facts about U.S. trade policy across this period. We focus on analyses using AVEs that combine all tariff types, including specific (per-quantity) tariffs and more complex compound and mixed forms. We show

[3] More recently, Bagwell, Staiger, and Yurukoglu (2020) digitize the bilateral negotiation records for the Torquay Round. Greenland and Lopresti (2024) compile product-level data for tariffs and imports for 1900-1930 and more aggregate import data for 1930-1940. Greenland, Lake, and Lopresti (2026a) and Greenland, Lake, and Lopresti (2026b) use product-level data for the Tokyo Round. Acosta and Cox (2024) compile product-level U.S. tariff commitments (but not import data) from the 1930s through the modern Harmonized Tariff Schedule, focusing on the ad valorem component of tariff rates as the authors note, "we have not consistently digitized unit values going back to 1930" (pg. 20). See Section 4.1 below for a discussion of the importance of specific tariffs during this time period.

[4] We use the most disaggregated product level that is available in the historical records: the approximately 5,500 product codes in the Schedule A (Census 1950). This is similar to the 7-digit TSUS level and approximately the 6-digit HS level. This paper refers to these product codes as Schedule A numbers.

that restricting attention to ad valorem tariffs alone is not advisable in this period: only about 39 percent of products have positive ad valorem tariffs,[5] and ignoring the specific component of compound and mixed tariffs—nearly 20 percent of the data—introduces non-random mismeasurement.

We confirm that U.S. tariffs decreased significantly for many products between Smoot-Hawley and the first GATT round, and that the bilateral agreements flowing from the 1934 Reciprocal Trade Agreements Act (RTAA) account for roughly half of the liberalization achieved between 1930 and 1947. Concessions from these bilateral agreements were extended on a most-favored-nation (MFN) basis to all countries except Cuba—a policy later applied to GATT concessions as well.[6]

We find no evidence that U.S. negotiators systematically substituted ad valorem for specific tariffs during this period, despite inflation having substantially eroded the protection afforded by specific tariffs set in 1930. This is surprising given that roughly half the items in our data set have a specific-tariff component, and specific tariffs are much less common in modern U.S. tariffs.

We then use this data set to test whether the TOT results hold for the 1930s and 1940s liberalization experience of the U.S. as it was building the global cooperative trading system. Specifically, we ask whether tariffs decreased more between Smoot-Hawley and the first GATT round for products with larger initial U.S. imports—that is, whether the efficient tariff is lower when non-cooperative imports are larger, conditional on the level of non-cooperative tariffs. Our data allow us to test the TOT theory in a setting difficult to obtain in modern data: a large trading country moving from unilateral to cooperative tariff setting. To the best of our knowledge, we are the first to provide such evidence for a large country, let alone for the largest trading country in the world before the consolidation of the GATT.

The regression results provide strong support for the role of importer market power,

[5] About 16 percent of products are duty-free and could be characterized as having an ad valorem tariff of zero. We discuss duty-free products in more detail in Appendix 10.3.1.

[6] Page XX of the 1950 Schedule A states "All of the currently operative reduced or modified rates, including those shown with a particular country or countries, are applicable to commodities, the growth, produce, or manufacture of all foreign countries with the following exceptions..." Those exceptions were for Cuba and the Philippines (Census 1950).

with effects larger than in the previous literature, which has mostly focused on developing economies acceding to the WTO (e.g., Bagwell and Staiger 2011). Conditional on Smoot-Hawley tariffs and industry fixed effects, GATT tariffs are lower when the U.S. imports more of the product. The baseline OLS estimates imply that a one-standard-deviation increase in 1933 U.S. import values corresponds to a 2.4 percent decrease in the average GATT tariff. This relationship is robust to alternative specifications, including using a Tobit model to address zero tariffs, separating trade liberalization that occurs through bilateral trade agreements versus the GATT, and replacing the import value variable with a measure of trade elasticities.[7]

Beyond the size of the country studied, our U.S. interwar context differs from modern tests of TOT theory (e.g., Bagwell and Staiger 2011) along three dimensions: (1) liberalization occurred in the absence of a cooperative global trading system; (2) negotiators faced distinctive political constraints; and (3) the changes were dynamic, spanning more than a decade. Perhaps most notably, we are looking at the beginning of the process of building the cooperative global trading system, whereas studies using modern data look at its culmination. We argue these constraints on internalizing the TOT externality do not alter the theory's predictions but make its results harder to detect, so detecting them in this setting was by no means a foregone conclusion.

Our work is related to several branches of the literature. Our TOT analysis is most closely related to Bagwell and Staiger (2011), who study the movement from non-WTO to WTO tariffs for 16 of the 21 countries that acceded to the WTO between 1995 and 2005. We extend their empirical approach to the case of the U.S. transition from Smoot-Hawley to the GATT. Our estimation results provide evidence that the TOT motive may have been stronger for the U.S. over this historical period than the sample of WTO-acceding countries from 1995 to 2005, as considered by Bagwell and Staiger (2011). Irwin and Soderbery (2021) also examine U.S. tariffs around Smoot-Hawley and highlight the importance of TOT and political-economy motives, but focus on the optimal unilateral tariff rather than the post-Smoot-Hawley change.

[7] We estimate demand and supply elasticities at the 4-digit level for import data from 1935 to 1938 by building on the Soderbery (2015) methodology. See Section 6.5 for details.

Several papers study the Smoot-Hawley period itself. Irwin (1998) separates the tariff's direct impact from the effect of price changes on specific-tariff incidence, and Bond et al. (2013) use product-level data from 1930 and 1933 to estimate productivity effects. On the trade-flow side, Mitchener, O'Rourke, and Wandschneider (2022) document the retaliatory trade war and its toll on U.S. exports, while Mitchener and Pedemonte (2026) use newly digitized monthly product-level data to estimate the short-run effects of the Smoot-Hawley increases on import prices and quantities, finding that affected imports collapsed and that the tariff's incidence fell almost entirely on U.S. importers. Acosta and Cox (2024) analyze the origins and persistence of regressivity in the U.S. tariff code. We complement these by examining how the Smoot-Hawley tariffs were dismantled, rather than how they were imposed or how they affected the economy.

A large qualitative literature documents the history of U.S. trade policy and the GATT, including comprehensive histories (Irwin 2017; Hoda 2018), accounts of individual GATT rounds (Evans 1971), studies of the USITC (Dobson 1976; Rosengarten, Summers, and Butcher 2017), legal treatments (Mavroidis 2016), and political-economy accounts of its origins (Destler 2005; Goldstein and Gulotty 2014; Irwin 2020). Our panel dataset also extends a tradition of data- and concordance-creation work (Feenstra 1996; Feenstra, Romalis, and Schott 2002; Pierce and Schott 2012) to a historical period for which such data have not previously been available.

The rest of the paper is organized as follows. The next section briefly describes the historical context, while Section 3 provides details on our data. Section 4 includes summary statistics for this new data set and stylized facts. Section 5 presents the TOT analysis, including the theoretical foundations, empirical strategy, and regression results. Robustness checks follow in Section 6. Finally, Section 7 concludes.

2 Historical Context

We examine the tariff liberalization undertaken by the United States between the U.S. Tariff Act of 1930 and the first round of the GATT in 1947 (i.e., the Geneva Round).[8] This section

[8]Sometimes the Geneva Round is referred to as Geneva I or Geneva 1947 because the later GATT round that was completed in 1956 was also named for Geneva. In this paper, reference to the Geneva Round is to

provides a brief historical background of U.S. trade policy between 1930 and 1947.

2.1 From the Smoot-Hawley Tariffs of 1930 to the Reciprocal Trade Agreements Act of 1934

In the midst of the Great Depression, U.S. President Herbert Hoover's proposal to increase tariffs to protect the agricultural sector turned into a large and general unilateral upward revision of the U.S. tariff schedule in the Tariff Act of 1930, known as the Smoot-Hawley Act. President Hoover intended that tariffs would be reduced under the flexible tariff provisions of Smoot-Hawley. However, regret at the economic damage wrought by the tariff increases and frustration with the cumbersome and time-consuming process for reducing those tariffs led to calls for a change (Dobson 1976).

That change came in the form of the Reciprocal Trade Agreements Act of 1934 (RTAA).[9] Framed as an amendment to the Smoot-Hawley Act—as would be all tariff bills until 1962—the RTAA authorized the President of the United States to change tariff rates by as much as 50 percent with only the need to notify the public of his intentions. This delegation of tariff setting powers to the president meant that the Smoot-Hawley Act was Congress's last general revision of U.S. tariffs.

2.2 1934–1946: The Era of U.S. Bilateral Trade Agreements

Between 1934 and 1946, the U.S. concluded a series of bilateral trade agreements that became the model for the GATT negotiations. The goal was to reduce tariffs, import quotas, export restraints, and other trade barriers to encourage trade and investment. Twenty seven nations signed bilateral agreements with the U.S. and nearly 2,000 U.S. products were impacted, accounting for about half of all items that were dutiable as of 1930. Canada even negotiated two agreements during this time with the U.S., one in 1936 and one in 1939. All commitments except those with Cuba were extended as MFN tariffs—that is, to all trading partners

the first GATT round in 1947. Although the Geneva Round tariffs went into effect on January 1, 1948, we follow the convention in the literature of referring to them as the Geneva 1947 tariffs.

[9] A short summary of the RTAA is available on the USTR website: https://ustr.gov/about-us/policy-offices/press-office/blog/2014/June/Eighty-years-of-the-Reciprocal-Trade-Agreements-Act.

regardless of whether they had trade agreements with the U.S.[10] We have recorded the tariff commitments with each country in our panel data except for those in the agreement with Cuba (where the commitments were non-MFN) and the two agreements with Nicaragua and Czechoslovakia that were terminated shortly after entry into force. The results of all these bilateral agreements are consolidated in the 1946 tariffs, which is the year prior to the first GATT round. Some products were included in multiple negotiations leading to multiple changes in tariff rates throughout this period of bilateral negotiations.

Interestingly, of the 23 contracting parties to the original GATT in 1947, eight had bilateral trade agreements with the U.S. in this era. Six of these are in our data: Belgium, Brazil, Canada, France, the Netherlands, and the United Kingdom. However, the other two, Czechoslovakia and Cuba, are excluded from our data, as previously discussed. This means that fourteen contracting parties to the GATT did not have a bilateral trade agreement with the U.S.[11]

2.3 The First GATT Round: Geneva 1947

Responding to events both at home and on the world stage, the U.S. was arguably the driving force behind the GATT. For the first GATT round in 1947, the U.S. created 28 country subcommittees to formulate tariff cut proposals based on the principal supplier rule. One of the top three suppliers of a commodity would be offered a tariff cut in return for reciprocal liberalization for U.S. goods. Then, the unconditional MFN principle would be applied to any tariff concession so that it was automatically and freely extended to all other suppliers, except a few communist countries (Dobson 1976).

By the mid 1940's, the U.S. had already made the RTAA-authorized maximum allowable reduction of 50 percent for about 42 percent of dutiable imports (Irwin 2017). President Truman, therefore, asked for additional authority to cut tariffs. In the Trade Agreements Extension Act of 1945, Congress gave the president the authority to reduce any tariff by 50 percent of the January 1, 1945 rate. Irwin (2017) argues that this was the "high point

[10] The distribution of these negotiations by country can be seen in Supplemental Data Appendix.

[11] This includes: Australia, Burma, Ceylon, Chile, China, India, Lebanon, Luxembourg, New Zealand, Norway, Pakistan, Southern Rhodesia, Syria, and South Africa.

of legislative support for the trade agreements program," facilitated in part because supply shortages—not import competition—were the focal issue amidst World War II.

With this negotiating authority, President Truman's administration instigated multilateral trade negotiations in Geneva, Switzerland starting in April 1947. The talks lasted until October 19, 1947 and produced the GATT. As U.S. legislation only required the president to give public notice of his intention to negotiate a trade agreement and then to "seek information and advice," the U.S. made its concessions effective as of January 1948 by executive order (Dobson 1976). The GATT was purely a trade agreement, not a treaty or an organization; thus instead of members, it had "contracting parties" (Irwin 2017).

The negotiations in Geneva were multilateral in principle. Still, the way they were carried out was similar to how the U.S. had operated since 1934 under the RTAA: a team from each country negotiated bilaterally with the representatives of several other countries on a product-by-product basis using the principal supplier rule (Irwin 2017). Once these bilateral talks were complete, each country created a single, consolidated schedule of tariff concessions that would be granted to all other contracting parties on an MFN basis.[12] These schedules were annexed to the GATT.

3 Historical U.S. Data

Our main data set combines detailed U.S. tariff commitments with corresponding product-level import data, consistently organized using the United States' Schedule A classification.[13] Following the Schedule A nomenclature, we treat each 7-digit code as a "product."[14] We begin by collecting statutory tariff rates for each product from contemporaneous government reports and, using a manual concordance, construct a product-level panel of tariffs spanning

[12] The U.S. had three sets of applicable tariffs at this time: the unilateral preferences given to the Philippines, the bilateral trade agreement concessions to Cuba, and the set of tariffs that were applied to all other countries. We focus on this last set of tariffs—which include concessions made in all trade agreements except the one with Cuba—throughout the paper. See Page XX of the August 1, 1950 Schedule A for more details.

[13] See Section 6.5 for a discussion of our product-country data for 1935-1938. Additional details on sources and construction are provided in Data Appendix 10.1.

[14] This level of disaggregation is comparable to the 6-digit Harmonized System (HS); for example, the Schedule A product 1059 200 (code as of 1948) "Broken rice" maps to HS code 1006.40 "Broken rice." We provide a more detailed time-series example of the mapping and concordance in the Supplementary Appendix.

from Smoot–Hawley (1930), through the bilateral trade agreements (1935–1939 and 1942–1944), pre-Geneva (1946), and post-Geneva (1947/1948).[15] When there is no explicit revision for a product in a given year, we assume the most recently observed rate remains in force, yielding a balanced product-year tariff panel.

A key contribution of our data is to provide ad valorem equivalents (AVEs) for products whose tariffs are not purely ad valorem. To construct AVEs and terms-of-trade measures, we match the tariff panel to U.S. import data reported under Schedule A, thereby linking product-level tariff rates to import values and unit prices by year. For most of the period, we rely on the Department of Commerce's *Foreign Commerce and Navigation of the United States*, which reports import quantities, values, and—when available—statutory rates, duties collected, and implied AVEs.[16]

Only 38.8 percent of the products in our data set have a simple ad valorem component that is nonzero. 15.5 percent of products are duty-free under Smoot-Hawley and remain that way throughout the period. A further 25.4 percent have a pure specific tariff. Another 14.7 percent have a compound tariff, that is, they have both an ad valorem and specific component in a given year. Finally, 5.6 percent of products have a mixed tariff, meaning a tariff of one type is bracketed by minimum (and possibly also a maximum) rate of another type.

To our knowledge, this is the first product-level time series that jointly covers the complete U.S. tariff schedule and corresponding import data over the period from Smoot–Hawley to the Geneva 1947 round. In the following section, we use this data set to document new stylized facts on U.S. trade liberalization from 1930 through the first GATT negotiation in 1947.

[15]The Geneva Round concluded in October 1947 and the U.S. tariff commitments entered into force on January 1, 1948. To be consistent with the literature, we refer to these as Geneva tariffs. We pair these tariffs with 1947 import data when computing AVEs because 1947 is the last year of imports unaffected by the new tariff schedule.

[16]Because this series ends in 1946, we use the Census Bureau's *United States Imports of Merchandise for Consumption, Commodity by Country of Origin for 1947*.

4 Summarizing U.S. Tariff Liberalization from Smoot-Hawley to the GATT

4.1 There is a Gradual Reduction in Tariffs Over Time

Table 1 confirms a well-known stylized fact: that U.S. tariffs experienced a relatively large decrease between Smoot-Hawley and the first round of the GATT in 1947. We find that from the Smoot-Hawley tariffs in 1930 to Geneva 1947, both specific and ad valorem tariffs were cut by a third or more when considering only products that were dutiable in 1930.

We exclude all items that are either duty-free or that have complicated tariffs from Panels A and B of Table 1 to allow for a clear comparison between ad valorem and specific tariffs, covering about 39 and 28 percent of dutiable products respectively. Among ad valorem products, the mean tariff binding falls from 40.36 to 25.65 percent between 1930 and 1947 (medians from 35 to 20 percent). Among specific-tariff products, the mean falls from 45.21 to 27.48 cents (medians from 5 to 3 cents).

Panel C presents the same statistics but substitutes AVEs calculated using same-year unit values from the import data. Calculating AVEs allows us to include products with specific, mixed, and compound tariffs, increasing the sample of dutiable tariffs by 89 percent relative to ad valorem tariffs alone.[17] As we show in Section 4.4, the products excluded when AVEs are not calculated are not randomly distributed across product types, making it especially important to concord import data so these products can be included. The mean AVE tariff binding falls from 49.62 percent in 1930 to 25.22 percent in 1947, a 49 percent drop—larger than the declines for ad valorem (36 percent) and specific (39 percent) tariffs alone. Two factors drive this larger decline: compound-tariff products, which are excluded from Panels A and B, start with higher AVEs and fall by more than other types; and inflation between 1933 and 1947 mechanically reduces AVEs even absent any change in bindings.

Panel D isolates the contribution of negotiated liberalization by holding unit values constant at 1933 levels. The mean AVE still falls substantially—by 35 percent between Smoot-

[17] Focusing on the Geneva Round, we are able to calculate AVEs for 963 products with specific tariffs throughout the period, 149 with mixed tariffs, and 528 with compound tariffs, plus 169 products whose tariff form changed between Smoot-Hawley and Geneva.

Table 1: Summary Statistics for Ad Valorem, Specific, and AVE Tariffs

Panel A: Ad Valorem Tariffs (percent)

	Mean	Median	Min	Max	N
Smoot-Hawley	40.36	35	5	110	2103
1946	34.39	30	2.5	110	2103
Geneva	25.65	20	2.5	110	2103

Panel B: Specific Tariffs (cents)

	Mean	Median	Min	Max	N
Smoot-Hawley	45.21	5	0.00375	3000	1381
1946	32.1	4	0.00375	1500	1381
Geneva	27.48	3	0.00375	1500	1381

Panel C: Ad Valorem Equivalents (percent)

	Mean	Median	Min	Max	N
Smoot-Hawley	49.62	40	0.44	742.27	4151
1946	33.75	27.5	0.04	512.21	3934
Geneva	25.22	20	0.03	205.56	3969

Panel D: Ad Valorem Equivalents using 1933 Unit Values (percent)

	Mean	Median	Min	Max	N
Smoot-Hawley	49.62	40	0.44	742.27	4151
1946	40.81	33.33	0.44	742.27	4157
Geneva	32.03	25	0.36	742.27	4159

Notes: All four panels exclude non-dutiable items. Panel A includes products that have only an ad valorem tariff throughout the period. Panel B includes products that have only a specific tariff throughout the period and standardizes the units in which the specific tariffs are expressed wherever possible. Panels C and D include products with all tariff types, but exclude products for which unit values, and hence AVEs, cannot be calculated. This impacts about 700 products whose tariff has a specific component but for which there were zero imports and for about two dozen products with positive imports for which no dollar value of imports is reported. Panel D holds unit values constant at 1933 levels to isolate the contribution of negotiated liberalization from inflation.

Hawley and Geneva—confirming that the bulk of the decline reflects genuine policy change rather than price-level effects.

4.2 Bilateral Trade Negotiations are Important, but the GATT is Faster

Regardless of which tariff measure above is used, what is clear is that the U.S. was active in negotiating bilateral agreements prior to the start of the GATT negotiations. Across all four measures presented in Section 4.1, at least half, if not more, of the reduction in U.S. MFN tariffs between Smoot-Hawley and Geneva 1947 is directly attributable to the bilateral trade agreements; this is because the U.S. converted all its bilateral concessions into MFN tariffs.[18] In Panel D of Table 1, the reduction between Smoot-Hawley and 1946 is about 18 percent, which is not that different from the 21 percent reported in Irwin (2017, p. 485) that uses more aggregated tariff data. Thus, bilateral negotiations between 1934 and 1946 made an important contribution to the statutory liberalization of this era.

Two related, but distinct measures shown in Figure 1 provide a first glimpse into the breadth of negotiations. The (red) dots show how many items with non-zero tariffs in 1930 saw new commitments in each year.[19] Geneva 1947 saw the largest number of items with commitments of any year: 2,302. The trade agreement with Ecuador in 1938 saw the fewest new commitments at just one out of the seventeen items on which the U.S. made commitments in this trade agreement; that is, the other sixteen items had already seen a reduction in a previous trade agreement.

The dark (blue) line indicates the number of items that had no negotiated change from Smoot-Hawley at each point in our data. For context, the light (orange) line visualizes the mean AVEs using the 1933 unit values as in Panel D of Table 1. This underlines the fact, discussed above, that while the magnitude of the reductions in AVEs were generally limited by statute, there was no such restriction on the number of products on which tariffs could be cut.

[18] Only the trade agreement with Cuba did not have its commitments extended in this way. The U.S. also gave the Philippines non-agreement preferential tariffs and did not extend these as MFN tariffs either.

[19] There were 1,419 items that were duty-free throughout the period or for which we do not have import data to calculate AVEs.

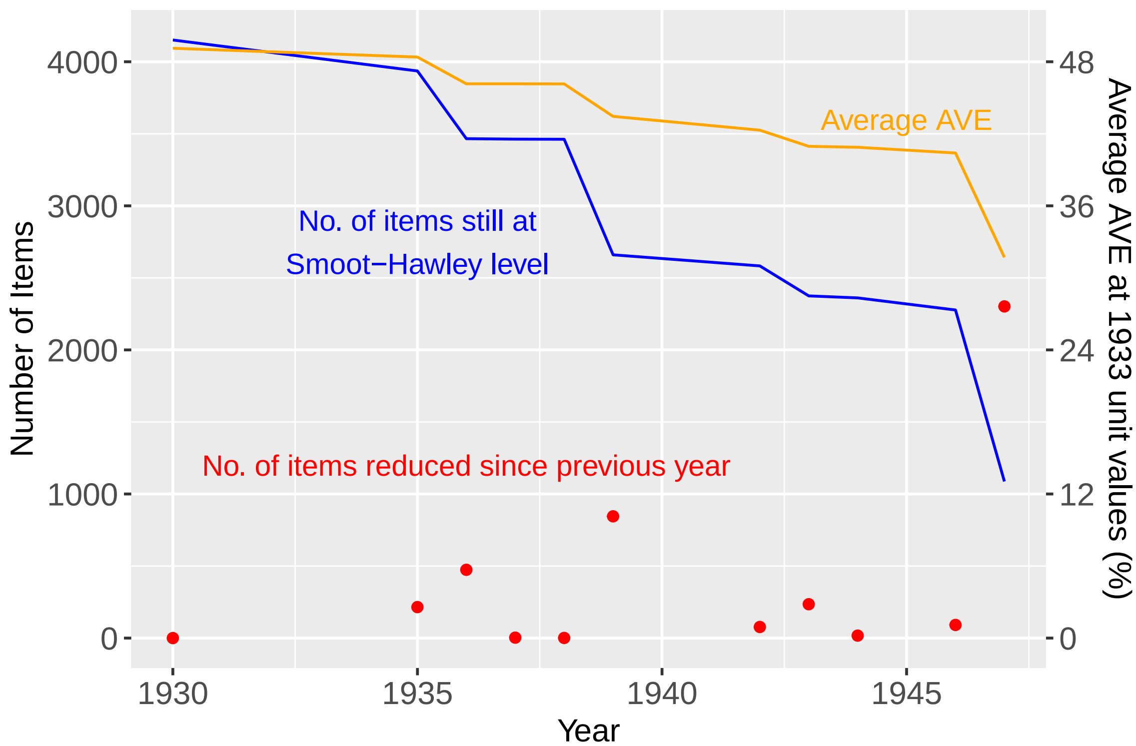

Figure 1: Tariff Reductions: Number of Items and Average Tariffs

Notes: The left axis shows the number of items that were reduced in each year (red) and a linear interpolation of the total number of items that remained at their 1930 level. The right axis shows a linear interpolation of the average AVE across all items in each year, holding unit values at 1933 levels. No bilateral trade agreements came into force in 1931-1934, 1940, 1941, 1945 or 1946. The small number of changes indicated by the red dot above 1946 are for unilateral tariff changes.

From another perspective, Figure 1 also visualizes the speed of liberalization. The steeper fall in the end of the period helps to illustrate what we see in the tables above: the GATT accomplished roughly the same amount of liberalization in six months as the bilateral trade agreements did in nearly a decade. This demonstrates the power of what both practitioners and scholars argue was the key feature of the GATT: the way in which it multilateralized the previous U.S. liberalization process (Bagwell, Staiger, and Yurukoglu 2020).

4.3 Some Products had Tariffs Reduced Repeatedly

The gradual reduction in tariffs we observe at the aggregate level is a product-by-product phenomenon. As there were institutional constraints on how far tariffs could be cut in a given time period, some products saw their tariffs reduced up to three times in our time series (see

Table 2). In Section 6.4 below, we discuss the possibility that the Congressional rule that tariff reductions must not exceed 50 percent constrained the degree of tariff reductions, and could have prevented the full internalization of the TOT externality. Table 3 shows that negotiated products that hit at least one of the 50 percent caps (from 1934 to 1946, then again starting in 1946) had higher average quantities, values, and unit values than those that did not.

Table 2: Multiple Tariff Reductions

Category	**Count**	**Composition**
No reductions	1,827	
Non-dutiable	871	Products with zero Smoot-Hawley tariffs
Dutiable	956	Products with positive Smoot-Hawley tariffs
One reduction	1,798	
Bilateral only	635	35.3% of one-reduction products
GATT only	1,163	64.7% of one-reduction products
Two reductions	976	
Both bilateral	41	4.2% of two-reduction products
Bilateral + GATT	935	95.8% of two-reduction products
Three reductions	33	All have one reduction in Geneva round
Total products	**4,634**	

Note: This table presents the distribution of tariff reductions across all products in the analysis data set; that is, it drops products for which some TOT regression variables are missing.

Table 3: Impacts of 50% Negotiating Caps: Average unit values, quantities, and values

Trade Agreement Status	**1933 UV**	**1933 Quantity**	**1933 Value**	**Count**	**Pct**
No reduction	283.5	3,044,052	98,488	942	25.0
Unconstrained reduction	179.1	1,354,510	97,458	1,448	38.5
Constrained by 50% cap	52.2	5,790,770	182,225	1,373	36.5

Note: Unit values are measured in US cents. Quantities are measured in each product's designated unit. Value is measured in US dollars. We take 1933 to be the beginning of the period and average over all products in three categories: those that saw no reduction, those that saw a reduction (possibly several times) but never hit the 50% cap, and those that hit at least one of the two 50% caps. Products that were constrained by the caps had much larger average import values, quantities and unit values (UV) than products that were negotiated but unconstrained. We exclude products that had zero tariffs under Smoot-Hawley, as they could not be reduced further.

4.4 Tariffs Display Significant Heterogeneity Across Product Categories

The summary statistics above hide variation in the speed and magnitude of liberalization across products. In the Schedule A, products are arranged into eleven industry groups; Table 4 lists these groups and shows the distribution of tariff types across them.

These industry groups range in size from 211 items for Group 00 (Edible Animal Products) to 918 items for Group 3 (Textiles). There is also variation in the proportions of specific, ad valorem, compound, and mixed tariffs across groups.[20] Only five items in Group 7 (Machinery) have a specific component, while all items have an ad valorem component. On the other hand, Group 00 (Edible Animal Products) has only 63 of its 211 items without a specific tariff component. Three groups have fewer than three percent of items with compound tariffs, while 30 percent of items in Group 4 (Wood and Paper) have compound tariffs.[21] The group with the largest number of compound tariffs is Group 9 (Miscellaneous), which has 192 items with compound tariffs. This heterogeneity in tariff form, and the fact that different units for specific tariffs tend to be concentrated in different groups makes it essential that we make comparisons among groups based on AVEs.

Table 5 presents the mean AVEs for these groups from Smoot-Hawley and Geneva. We also include AVEs for 1939 and 1946 to provide some additional insights into the impact of bilateral trade agreements prior to the GATT. We continue to use constant 1933 unit values for these AVEs.

Tariff liberalization through bilateral agreements and the Geneva round across the industry groups was broad-based. For example, every group saw a reduction in AVEs of at least five percent from Smoot-Hawley until 1939, and at least ten percent from Smoot-Hawley until 1946.[22] Across the industry groups, the magnitude of the average tariff decrease varies. Between Smoot-Hawley and the Geneva round, Group 8 (Chemicals) and Group 9 (Miscellaneous) experienced tariff decreases around 26 percent, while Group 1 (Edible Vegetables)

[20] Here, we include non-dutiable items in the ad valorem category.

[21] 189 items switch tariff type at some point. The counts in Table 4 use the tariff form under Smoot-Hawley. See Section 4.6 for more details.

[22] Bown and Irwin (2017) present summary data from the USITC on the reduction in rates from Smoot-Hawley to just before Geneva and then to the Geneva rates using aggregated data.

Table 4: Industry Groups in Schedule A

Group	Title	Count of Tariff Type				Total Items
		AV	Specific	Compound	Mixed	
00	Animals and animal products, edible	63	110	5	23	211
0	Animals and animal products, inedible	325	24	29	76	460
1	Vegetable food products and beverages	142	353	5	1	503
2	Vegetable products, inedible	313	97	16	2	437
3	Textile fibers and manufactures	587	119	172	30	918
4	Wood & paper	205	38	105	0	350
5	Nonmetallic minerals	276	124	88	19	516
6	Metals and manufactures	361	249	119	6	758
7	Machinery and vehicles	217	0	5	0	227
8	Chemicals and related products	223	184	85	1	510
9	Miscellaneous	340	141	192	1	680

Notes: The Schedule A groups products into eleven industry groups. Both the number of products and the types of tariffs vary widely across groups. The counts here are for the tariff type (AV for Ad Valorem, Specific, Compound, and Mixed) under the Smoot-Hawley Act in 1930. When the counts of the four tariff types do not add up to the total, it is due to products that did not exist in 1930; there are 99 such products across all groups. Changes in tariff type by 1947 are discussed in the section on tariff form switching below.

had a decrease in the mean tariff of over 40 percent.

Table 5: Mean AVE Tariffs (1933 Unit Values) by Industry Groups, Smoot-Hawley to Geneva

Group	AVEs				Percentage Decrease		
	SH	1939	1946	Geneva	SH to 1939	SH to 1946	SH to Geneva
00	34.3	30.0	26.9	20.3	12.5	21.5	40.9
0	27.4	23.4	22.4	16.9	14.4	18.3	38.2
1	52.9	47.8	36.8	30.9	9.6	30.5	41.7
2	19.5	16.9	16.0	12.4	13.7	18.0	36.4
3	59.7	52.5	49.7	36.4	12.1	16.7	39.0
4	23.8	20.6	19.9	15.8	13.5	16.3	33.7
5	36.8	33.5	30.3	23.6	9.1	17.8	36.1
6	43.9	39.3	38.2	29.5	10.4	12.8	32.7
7	28.1	23.2	22.9	16.8	17.4	18.5	40.3
8	31.5	29.7	28.2	23.2	5.7	10.5	26.4
9	53.4	46.3	45.6	39.5	13.4	14.7	26.1

Notes: The Schedule A groups products into eleven industry groups and the rate of tariff reduction varies across these groups. The second through fifth columns give the mean tariff by group in June 1930 (SH stands for the Smoot-Hawley Act of 1930), midway through the bilateral reductions in 1939, at the end of bilateral reductions in 1946, and after the Geneva (first) round of the GATT was concluded. The last three columns show the decreases from 1930 to 1939, 1946, and 1947 (Geneva).

4.5 Tariff Increases are Rare

Although it is rare, we see a few instances of tariffs increasing over the period 1930–1947. There are only 102 cases where the AVE of the Geneva tariff is higher than the AVE of the Smoot-Hawley tariff. Most of these are not due to statutory increases, but rather to reductions in the unit value from 1933–1947. These increases are scattered across all the groups except Group 7 (Machinery).

There was, however, a statutory tariff increase for 22 products. That is, for these products, the AVE of the Smoot-Hawley tariff is lower than the AVE of the Geneva tariff at 1933 unit values. Most of these increases are caused by unilateral U.S. actions such as Section 336 or the Sugar Act.[23]

4.6 Switching between Ad Valorem and Specific Tariffs is Also Rare

Given that inflation was reducing the incidence of specific tariffs during this period (Irwin 2017), one might have expected specific tariffs to be converted into ad valorem tariffs to avoid a loss of protection. Another reason we might expect this conversion is simply that only eight percent of U.S. tariffs were specific during the three decades since 1988 (Teti 2020).

However, from 1930 to 1947, the tariff for only 189 products changed form.[24] Surprisingly, the most common change was transforming an ad valorem tariff into a mixed tariff, that is, where a tariff of one type is bracketed by minimum and/or maximum rates of another type for a single product. This was the case for 97 products, about a third of which were concentrated in Group 3 (Textiles) while the rest were scattered throughout Groups 5–9.

Almost the same number of products moved out of the specific category, with 17 becoming

[23] See Appendix 10.3.2 for details on the special provisions. For two products—both types of fur hats—the increase is caused by a change in the form of the tariff combined with a low unit value. The tariff for Schedule A number 0754 000 changed from compound to specific with an ad valorem minimum, while the tariff for Schedule A number 0754 500 switched from compound to ad valorem. For one product, Schedule A number 9760 500—combs made of cellulose compounds—the tariff form (compound) did not change, but the values did. Under Smoot-Hawley, the specific part was 2 cents each and the ad valorem part was 35 percent; in Geneva, the tariff was 3 cents each plus 20 percent ad valorem with a 1933 unit value low enough to create an increase in the AVE.

[24] This excludes 99 products that did not exist in 1930.

mixed, 20 becoming ad valorem, and 6 becoming compound. All but three of these were in Group 5 (Nonmetallic Minerals) or Group 6 (Metals).

Of products that had compound tariffs under Smoot-Hawley, 36 shifted to mixed tariffs by 1947, while 10 more shifted to ad valorem. There was only one product that had a mixed tariff under Smoot-Hawley whose tariff form changed by 1947—to ad valorem. These changes were distributed fairly evenly across groups.

Thus, we see no clear pattern of moving toward or away from one type of tariff over this period. The shift from specific to ad valorem tariffs must therefore have occurred in later GATT rounds. Greenland, Lake, and Lopresti (2026a) find a significant number of specific lines remaining as late as the Tokyo Round.

5 Terms-of-Trade Analysis

The previous section provides an overview of the wide-ranging U.S. tariff liberalization following Smoot-Hawley through the first GATT round in 1947. There, we document the patterns of tariff reductions: broad-based tariff reductions, substantial cross-product heterogeneity, and several institutional constraints. This does not, however, explain what drove U.S. tariff changes over this period. We turn now to that question, focusing on the TOT motive as one possible explanation.

5.1 Conceptual Framework

The TOT theory posits that countries with importer market power face a temptation to set higher import tariffs to shift some of the cost of protecting domestic industries onto their trading partners. Thus, importer market power can lead to higher unilateral tariffs. The TOT theory suggests that trade agreements can allow for countries to achieve lower tariffs that can increase their welfare.

Bagwell and Staiger (2011) show the TOT theory can be used to define a relationship that predicts the level of negotiated tariffs from pre-negotiation information on tariffs (τ), import volumes (M), prices (p), and import demand (σ) and foreign export supply (ω^*)

elasticities. For a domestic product, the following expression can be derived:

$$\tau^{PO} = \beta_0 + \beta_1 \tau^{BR} + \beta_2(\sigma^{BR}/\omega^{*BR})(M^{BR}/p^{BR}). \tag{1}$$

The superscript PO denotes the politically optimal level for the domestic government after entering trade negotiations, and BR denotes the best-response level in the absence of trade negotiations (i.e., pre-negotiation). The authors establish that the TOT theory predicts that $\hat{\beta}_1 > 0$ and $\hat{\beta}_2 < 0$.

Moreover, Bagwell and Staiger (2011) demonstrate that, in the case of linear demand and supply, the TOT theory yields a more straightforward form of the above expression. In this case, the negotiated tariff cut should be larger in magnitude when the ratio of pre-negotiation import volume to world price (denoted m^{BR}) increases. This can be represented by the following equation:

$$\tau^{PO} = \beta_0 + \beta_1 \tau^{BR} + \beta_2 m^{BR}. \tag{2}$$

5.1.1 Barriers to Terms-of-Trade Internalization

The trade liberalization episode that we study in this paper differs from the canonical setting in the traditional TOT literature. Most notably, we focus on the beginning of the liberalization process instead of its culmination. Instead of operating within a well-established global trading system, the U.S. was taking the first steps to build the cooperative global trading system that was in place in the second half of the twentieth century.

Another defining characteristic of our context is the gradual implementation of tariff reductions over more than a decade, rather than through a single, comprehensive reform. This gradualism is reflected in both the breadth of product coverage and the depth of the tariff changes. We interpret this dynamic trajectory through the lens of Maggi and Rodriguez-Clare (2007): in the absence of constraints, the first reduction should internalize the terms of trade externality. In our context, full internalization was likely impeded by domestic institutional barriers and global market frictions. This implies that further rounds of reductions might still be efforts to complete the terms of trade internalization.

Among products that were negotiated, many products had their tariffs reduced more than once (Table 2). Drawing on our data and historical documents, we argue that such repeated reductions are mainly driven by the 50 percent cap mandated by the U.S. Congress, insufficient market access available from negotiating partners, and fluctuations in world prices during the relatively long period we study. All these factors could cause the negotiated tariff to deviate from the politically optimal tariff level.

The U.S. also faced significant constraints in determining which tariffs could be reduced. Rather than negotiating across the full range of products, reductions were limited by institutional rules—most notably, the principal supplier rule and the requirement of reciprocity. These provisions meant that U.S. negotiators could offer concessions only on products primarily exported by a limited set of trading partners. Even within this restricted set, not all products received tariff cuts. We hypothesize that this selective pattern likely reflects fixed costs associated with negotiating each individual tariff line, which constrained the scope of liberalization.

We argue that these key differences should be understood as constraints on the TOT internalization process. Thus, they serve to make the TOT hypothesis more difficult to detect but do not fundamentally alter the intuition or the conceptual framework presented above.

5.2 Empirical Strategy

To study the relationship between importer market power and the change in tariffs between Smoot-Hawley and the Geneva round, we extend the Bagwell and Staiger (2011) empirical strategy to the case of U.S. GATT and Smoot-Hawley tariffs with the following fixed effects regression model:

$$\tau_g^{GATT} = \alpha_G + \beta_1 \tau_g^{SH} + \beta_2 V_g^{1933} + \varepsilon_g, \tag{3}$$

where τ_g^{GATT} is the U.S. bound Geneva 1947 tariff on the 7-digit Schedule A product g and τ_g^{SH} is the U.S. bound Smoot-Hawley tariff. AVEs are used for this analysis to avoid challenges with separately comparing ad valorem and specific tariffs. V_g^{1933} is the U.S. import value in 1933. As we use U.S. import data, we follow the primary empirical approach used in Bagwell and Staiger (2011) and replace the m^{BR} variable in Equation 2 with U.S.

import values in Equation 3.[25] To account for the right skewness in the import data, we also consider replacing the level of U.S. import values with the log transformation and a binary variable that denotes import values above the median import value. The detailed trade data is at the tariff line level, so we infer that import values with no recorded import value and quantity are zero.

The variable α_G denotes the 93 industry fixed effects for Schedule A subgroups.[26] These subgroups are similar to 2-digit fixed effects when using the modern HS system; however, directly using the first 2 digits of Schedule A codes as fixed effects is not preferred as the Schedule A codes do not follow a rigorous hierarchical structure like HS codes do. These fixed effects allow for the comparison of 7-digit products within the same industry and control for industry-level characteristics that could impact U.S. trade policy, including the political economy environment for each industry. Using fixed effects to control for a wide range of observables and unobservables is particularly useful as there are limited historical data readily available to link to our data set. Finally, ε_g is the error term and we use OLS estimation with heteroskedasticity-robust standard errors.

In relation to the TOT theory, τ_g^{GATT} proxies for the efficient politically optimal tariffs, while τ_g^{SH} and V_g^{1933} proxy for the U.S. unilateral best-response tariffs and import values, respectively. As previously discussed, Bagwell and Staiger (2011) show that the TOT theory predicts that $\hat{\beta}_1 > 0$ and $\hat{\beta}_2 < 0$. In other words, conditional on the level of non-cooperative tariffs, the efficient tariff level should be lower when non-cooperative imports are larger. Moreover, the authors note that $\hat{\beta}_2$ approaches zero in the case of a small economy relative to the world market for a given product. Given the focus by Bagwell and Staiger (2011) on developing economies in their empirical application, we expect our $\hat{\beta}_2$ estimates to be larger in magnitude as we focus on the case of an economy that was the largest in the world during the period under study (Irwin and Soderbery 2021). Moreover, both conditions for the terms-of-trade mechanism are met here: Irwin and Soderbery (2021) estimate substantial U.S. importer market power and find the Smoot-Hawley tariff reflected terms-of-trade motives.

[25] In other words, we do not use U.S. unit values to infer world prices to calculate m^{BR} in Equation 2. In a robustness check, we use the more general specification in Equation 1.

[26] These groups are defined on page VII of the 1950 Schedule A in a table entitled "Import Commodity Group and Subgroup Code Designations."

We use 1933 import values for both V_g^{1933} and to compute AVEs of specific tariffs for τ_g^{SH}. We choose 1933 as the base year for three reasons.[27] First, using 1933 instead of 1930 gives U.S. imports a few years to adjust following the passage of Smoot-Hawley. Since we are interested in a measure of non-cooperative import values, import data must reflect trading conditions under the Smoot-Hawley tariffs. Second, 1933 was the last year of widespread deflation in the U.S., so this puts us near the trough in prices. Finally, 1933 was the last year before the RTAA in 1934, which authorized the beginning of trade negotiations following Smoot-Hawley. As previously discussed and a point we return to with a robustness check, the RTAA resulted in the U.S. signing bilateral trade agreements prior to the Geneva round.

5.3 Terms-of-Trade Summary Statistics

Table 6 includes the summary statistics for the variables in Equation 3 as well as those used in robustness checks we introduce below. The baseline sample of 4,634 products differs in two ways from the sample used for the summary statistics in Section 4. First, it excludes products that have missing values for any of the three key variables: Geneva AVE tariffs, Smoot-Hawley AVE tariffs, and 1933 import values. Second, it includes non-dutiable items. In the baseline sample, 872 of the Smoot-Hawley tariffs are already duty-free. As Smoot-Hawley tariffs were set unilaterally and the U.S. *could* have increased these tariffs, duty-free products in Smoot-Hawley that continued to be duty-free in the Geneva round are informative to understanding trade policy over this period. The inclusion of duty-free Smoot-Hawley tariffs is also consistent with the approach used by Bagwell and Staiger (2011), where the authors included pre-WTO duty-free products.

The mean Geneva tariff of 20.41 percent is well above duty-free treatment and is about half of the mean Smoot-Hawley tariff of 39.45 percent. As expected, this implies that studying U.S. tariff outcomes from the GATT is not the trivial case of predicting free trade for all products. To provide further insight into the distribution of tariffs, Figure 2 plots the kernel densities for Smoot-Hawley and Geneva AVE tariffs. There is a leftward shift in the distribution of tariffs in the Geneva round relative to Smoot-Hawley, showing tariff decreases

[27] We also try using smoothed unit values that combine multiple years in the 1930s and the estimation results are similar. These results are available upon request.

are prevalent across products.

Finally, the mean 1933 import value is about $280,000 and has a wide range of variation with a standard deviation of $2.82 million. The median import value is $14,119 and this threshold is used to define the binary variable of high 1933 import values (i.e., the binary variable equals to one when imports are above the median and zero otherwise). 464 of the 4,634 observations have a 1933 import value of zero, so the sample decreases to 4,170 observations when using the log of 1933 import values.[28]

5.4 Main Regression Results

Table 7 presents the results of Equation 3. Column 1 includes import values in millions of U.S. dollars. The estimate on Smoot-Hawley tariffs is positive and significant at the 1 percent

[28] Regarding zero import values, Bagwell and Staiger (2011) also notes that, "the terms-of-trade theory therefore also predicts that we should observe small negotiated tariff cuts on products where the importing country has raised its tariffs to near prohibitive levels, regardless of the foreign export supply elasticity that it faces" (pg. 1244). Prohibitive tariff levels imply that import values go to zero, so to provide some insights on this prediction, we compare the mean tariff cut between Smoot-Hawley and the Geneva round for products with zero imports in 1933 to those with positive imports in 1933. For products with zero imports, the mean tariff decrease is about 11 percentage points, while for positive imports the decrease is about 19.9 percentage points. Thus, as predicted by the TOT theory, products with zero imports face smaller tariff decreases, on average.

Table 6: Terms-of-Trade Summary Statistics

	Mean	Std. Dev.	Min	Max	N
Smoot-Hawley AVE Tariff (1933 Unit Values)	39.45	37.10	0	581.48	4,634
AVE Tariff Prior to Geneva (1946 Unit Values)	26.89	25.24	0	331.20	4,486
Geneva AVE Tariff (1947 Unit Values)	20.41	20.35	0	192.23	4,634
Geneva AVE Tariff (1933 Unit Values)	25.24	26.53	0	407.04	4,633
1933 Import Value	0.28	2.82	0	124.14	4,634
Log 1933 Import Value	-4.21	2.70	-13	4.82	4,170
High 1933 Import Value	0.50	0.50	0	1.00	4,634
η_g	0.05	1.34	0	41.09	2,012
Log η_g	-16.30	6.51	-37.12	3.72	1,890
High η_g	0.50	0.50	0	1	2,012

Notes: AVEs are listed as percentages. The sample size decreases slightly for the variable "AVE Tariff Prior to Geneva (1946 Unit Values)" due to the availability of 1946 import values and quantities used to calculate 1946 unit values. Import values are in millions of U.S. dollars. The "Log 1933 Import Value" variable drops observations that have an import value of zero. Following Equation 1, $\eta_g \equiv (\sigma_g^{BR}/\omega_g^{*BR})(M_g^{BR}/p_g^{BR})$, see Section 6.5 for more details.

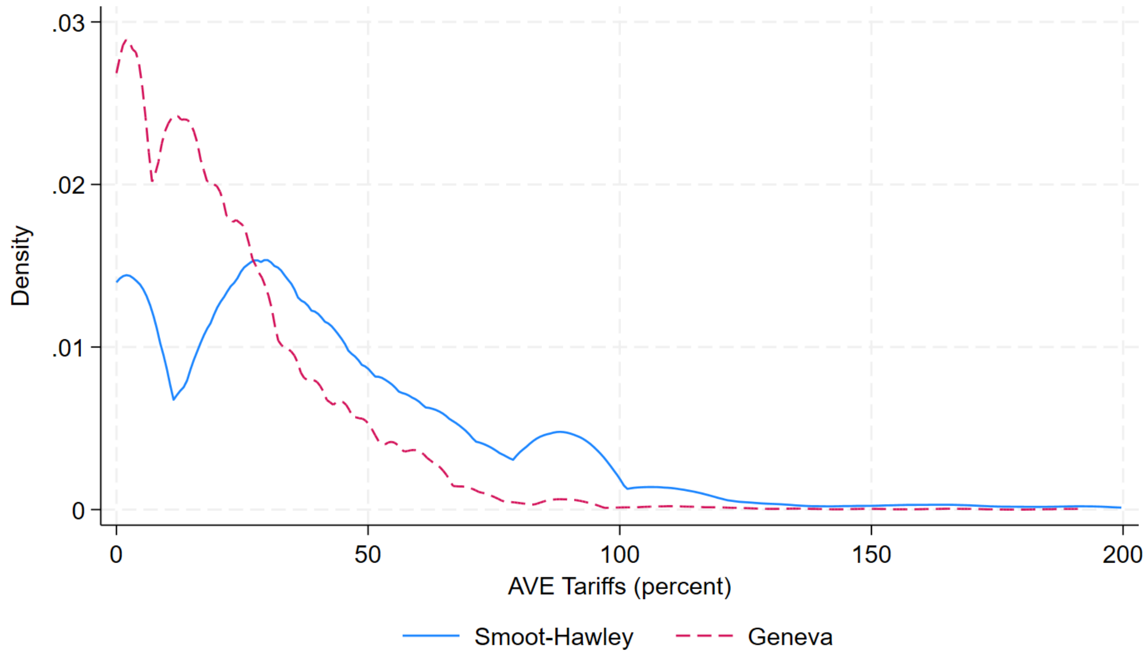

Figure 2: Distribution of Tariffs

Notes: For ease of exposition, several AVE tariffs above 200 percent are omitted from the figure. Smoot-Hawley AVE tariffs use 1933 unit values to calculate AVEs for observations that include specific tariffs. Geneva AVE tariffs use 1947 unit values.

level, while the estimate on import values is negative and significant at the 1 percent level. Both estimates are consistent with the TOT theory predictions and imply that, conditional on the Smoot-Hawley tariffs, Geneva tariffs are lower when the U.S. imports more of the product. Column 2 uses a log transformation of import values to account for the few larger import values relative to the overall sample. Using log imports results in observations with zero import values being dropped.[29] The estimate on log import values is also negative and significant at the 1 percent level. As predicted, the estimate on Smoot-Hawley tariffs continues to be positive and significant and is similar across columns 1–3. To further explore how larger import values influence the results, we use the binary variable of high 1933 import values that indicates whether imports are above the median import value. This estimate in column 3 is negative and significant, which is also consistent with the TOT motive.

[29]The number of observations decreases by one compared to the number of observations listed for log imports in Table 6, as one product is the only observation in an industry fixed effect grouping (i.e., a singleton observation) and it is dropped from the regression.

Table 7: Terms-of-Trade Baseline Results

Dependent Variable: Geneva AVE Tariff (1947 Unit Values)	OLS			Tobit		
	(1)	(2)	(3)	(4)	(5)	(6)
Smoot-Hawley AVE Tariff	0.295***	0.279***	0.294***	0.344***	0.320***	0.343***
	(0.022)	(0.024)	(0.022)	(0.008)	(0.008)	(0.008)
1933 Import Value	-0.169***			-1.273***		
	(0.061)			(0.305)		
Log 1933 Import Value		-0.526***			-0.745***	
		(0.094)			(0.103)	
High 1933 Import Value			-2.267***			-3.061***
			(0.413)			(0.507)
Observations	4,634	4,169	4,634	4,634	4,169	4,634
R-squared	0.545	0.549	0.547			
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors in parentheses (OLS are heteroskedasticity-robust). The log of 1933 import values drops observations that have an import value of zero. The Tobit model specifies zero as a lower limit of the dependent variable. Smoot-Hawley AVE tariffs use 1933 unit values to calculate AVEs for observations that include specific tariffs. Singleton observations are dropped. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

To quantify the economic significance of $\widehat{\beta}_2$ in Equation 3, we consider a one standard deviation change in the import values relative to the mean GATT tariff. Using the point estimates in column 1, a one standard deviation increase in import values, all else equal, reduces bound GATT tariffs by about 0.48 percentage points ($-0.169 \times 2.82 = -0.4766$). The mean GATT tariff is 20.41 percent, which implies about a 2.4 percent decrease in the average GATT tariff (0.48/20.41 = 0.024). In comparison, this is larger than the overall result from Bagwell and Staiger (2011), where their estimates imply a 1.7 percent decrease in the average bound WTO tariff. To interpret the remaining import value variables, the point estimate on the log import value in column 2 implies that doubling US import values, all else equal, reduces bound GATT tariffs by about 0.53 percentage points. The column 3 estimate on the binary import value variable implies that high levels of US import values, all else equal, reduces bound GATT tariffs by about 2.27 percentage points.

About 19 percent of the Geneva tariffs in our sample (or 860 observations) are duty-free. If disturbances are normally distributed and homoskedastic, this suggests that a Tobit model may be preferred over the OLS model. The Geneva tariffs are technically not censored, but have a lower bound of zero as tariffs are not negative. Nevertheless, we use a Tobit model as one robustness check and report the results in columns 4–6.

The Tobit results provide additional support for the TOT theory predictions.[30] The point estimates on the import variables continue to be significant and are more negative than the OLS results. In column 4, the magnitude of the import value estimate increases by over seven fold, while in columns 5 and 6 the increase in magnitude is more modest at about 42 percent and 35 percent, respectively. However, the Tobit point estimates are not the best measure to compare to the OLS coefficients, so we also compute the average marginal effect of the change in 1933 import values conditional on positive tariffs for the Tobit model. We find that the adjusted Tobit coefficients are larger in absolute value than the OLS estimates, yet smaller than the Tobit point estimates.[31] We view the Tobit results as evidence that accounting for the non-negativity constraint on tariffs provides further support for the TOT theory predictions.

Thus, we find strong support for the TOT hypothesis *despite* the numerous barriers to TOT internalization discussed in Section 5.1.1. We next turn to demonstrating the robustness of these results.

6 Robustness Checks

6.1 Changes in Unit Values Over Time

As previously discussed, AVEs for specific tariffs could increase or decrease over time even if the tariff commitments are unchanged because of changes to unit values. For example, higher unit values due to inflation would lead to smaller AVEs, all else equal. To explore the impact of unit value changes, we check whether replacing the Geneva AVEs at 1947 unit values with AVEs that hold unit values constant using 1933 unit values impacts our results.

In addition to Geneva AVEs calculated with 1947 unit values, Table 6 includes summary

[30] In all robustness checks below, we include OLS and Tobit estimation results to highlight that the general findings are supported by both models.

[31] Results are available upon request.

statistics when holding unit values constant at 1933 unit values.[32] The mean Geneva AVE tariff increases to 25.24 percent, up from 20.41 percent when using 1947 unit values, but is still well below the mean Smoot-Hawley tariff of 39.45 percent. This suggests that AVEs using time-varying unit values may overstate the negotiated tariff decreases between Smoot-Hawley and the Geneva round. However, if policymakers are actively aware of the changes in unit values—through inflation or deflation in the economy—and account for this when negotiating specific tariffs, then the calculated AVEs with time-varying unit values could be the tariff measure of interest in trade policy negotiations. We do not take a stance on whether holding unit values constant is preferred; instead, we present a robustness check that holds unit values constant.

To provide insights into the implications of changes in unit values, Figure 3 plots the relationship between Geneva and Smoot-Hawley AVEs (the solid blue circles) and this same relationship when holding unit values constant at 1933 unit values (the black empty circles). Most striking is that there is more variation in the Geneva AVEs using time-varying unit values. This is expected given that changes in these AVEs from Smoot-Hawley to Geneva encompass both unit value changes that impact AVEs for specific tariffs and changes in the tariff commitments. In contrast, when holding unit values constant, AVEs only include changes in the tariff commitments. Points along the 45-degree line are products with Smoot-Hawley AVEs equal to Geneva AVEs, with points below (above) this line representing tariff decreases (increases) over the time period. While tariff increases between Smoot-Hawley and Geneva are rare, 102 products have Geneva AVEs that are larger than the Smoot-Hawley AVEs when using time-varying unit values, and only 22 products have Geneva AVEs larger than Smoot-Hawley AVEs when using 1933 unit values.[33]

Table 8 includes the regression results when the dependent variable in Equation 3 is the Geneva AVE tariff that uses 1933 unit values. Across the columns, the coefficient estimates on the Smoot-Hawley tariffs are larger than in Table 7. In contrast, the magnitude of the point estimates on the level, log, and binary import value variables are generally smaller, but continue to provide negative estimates that are consistent with the TOT motive. Only

[32] AVEs that hold unit values constant are labeled with "(1933 Unit Values)" in Table 6.

[33] See Section 4.5 above for details.

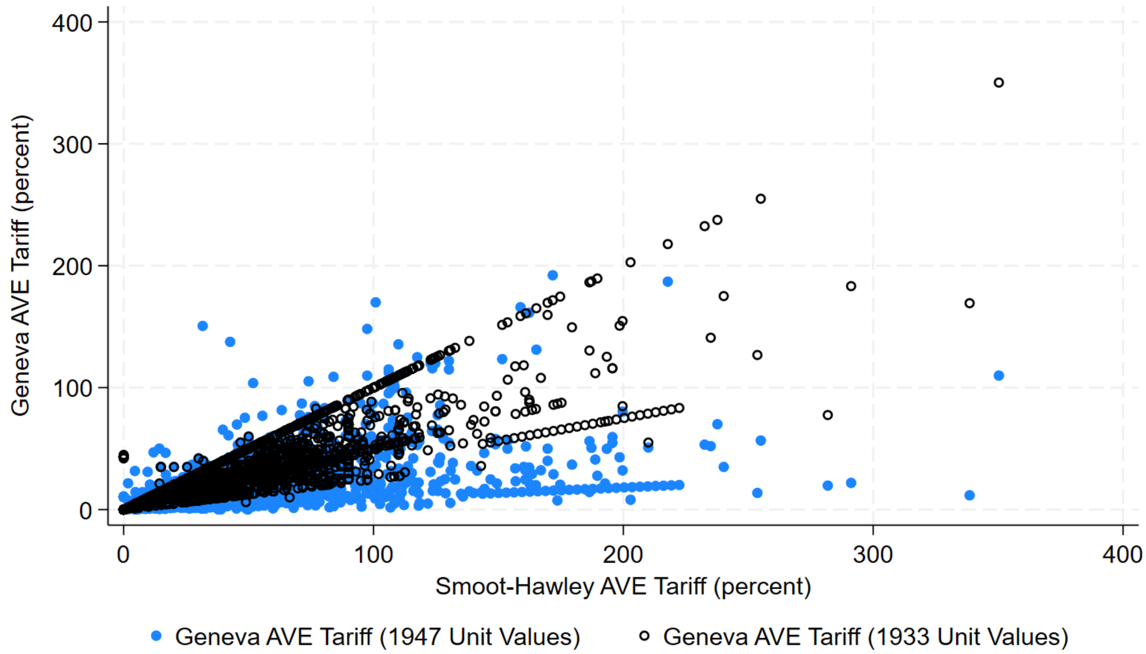

Figure 3: Time-varying Unit Values Versus Constant 1933 Unit Values

Notes: For ease of presentation, this figure omits one product with a Smoot-Hawley AVE above 400 percent. Geneva AVE tariffs differ in the unit values (from either the year 1947 or 1933) used to calculate AVEs for observations that include specific tariffs. Smoot-Hawley AVE tariffs use 1933 unit values.

the level import value estimate in column 1 is insignificant; however, the preferred log and binary import variables continue to be statistically significant for OLS estimation, and all three functional forms with the Tobit results are significant.

The smaller and less significant results using the 1933 unit values is suggestive evidence that negotiators were focusing on real-valued tariffs instead of nominal ones. According to the TOT theory, tariffs should be set to achieve optimal domestic prices given world prices. In the absence of product-level data for world prices, our baseline specification uses 1947 U.S. import unit values as a proxy. If negotiators were focusing on real-valued tariffs instead of nominal ones, then the 1947 unit values are a better proxy than the 1933 unit values for most products.

Overall, the estimates in Table 8 suggest that while the changes in unit values impact the magnitude of the estimated effect, holding unit values constant at 1933 levels yields regression estimates that remain consistent with the TOT theory predictions.

Table 8: Robustness Check, Holding Unit Values Constant at 1933 Levels

Dependent Variable: Geneva AVE Tariff (1933 Unit Values)	OLS			Tobit		
	(1)	(2)	(3)	(4)	(5)	(6)
Smoot-Hawley AVE Tariff	0.638***	0.662***	0.637***	0.692***	0.707***	0.691***
	(0.018)	(0.019)	(0.018)	(0.007)	(0.007)	(0.007)
1933 Import Value	-0.053			-1.325***		
	(0.048)			(0.285)		
Log 1933 Import Value		-0.220***			-0.434***	
		(0.078)			(0.088)	
High 1933 Import Value			-0.792**			-1.462***
			(0.370)			(0.446)
Observations	4,633	4,168	4,633	4,633	4,168	4,633
R-squared	0.788	0.806	0.788			
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors in parentheses (OLS are heteroskedasticity-robust). The log of 1933 import values drops observations that have an import value of zero. The Tobit model specifies zero as a lower limit of the dependent variable. Smoot-Hawley AVE tariffs use 1933 unit values to calculate AVEs for observations that include specific tariffs. Singleton observations are dropped. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6.2 Bilateral Trade Agreements

As discussed in Section 2, reductions in U.S. MFN tariffs between 1930 and 1947 come from both the Geneva negotiations and a series of bilateral trade agreements. The tariff commitments from these trade agreements were extended on an MFN basis by the U.S. This implies that observed Geneva round tariffs capture tariff changes not only from GATT negotiations but also tariff changes from previous bilateral trade agreements.[34]

From one point of view, applying bilateral trade agreement tariffs on an MFN basis is not entirely consistent with the TOT theory since the U.S. is only exchanging tariff concessions with trade agreement partners but extending those tariff concessions multilaterally.[35] At the same time, these bilateral negotiations—although not taking place simultaneously—were linked together in a way that is similar to those in the GATT through the extension of

[34]Tariff changes prior to GATT also include changes due to other, non-trade agreement provisions, a point we return to in a later robustness check.

[35]One limitation of our data is we are not able to observe whether other non-member countries were also reducing their tariffs as the U.S. was signing these trade agreements.

MFN. In addition, having trade liberalization prior to GATT is helpful to our empirical approach as it supports that the U.S. was making a significant shift from a non-cooperative to cooperative tariff setting—which would probably be less likely to occur if there was only one opportunity for trade negotiations.

Since Geneva-round tariffs in 1947 encompass a series of negotiations over many years, we present two robustness checks in Table 9 that disentangle the tariff changes from the bilateral trade agreements from the tariff changes in the Geneva round. First, we substitute 1946 AVE tariffs as the dependent variable in Equation 3. 1946 is the last year before the GATT, and it includes all the bilateral tariff changes that occurred between Smoot-Hawley and 1946. The summary statistics in Table 6 include the 1946 AVEs (labeled "AVE Tariff Prior to Geneva (1946 Unit Values)") and the mean tariff is about 26.9 percent. This suggests that the U.S. tariff changes prior to GATT account for about 66 percent of the average tariff change between Smoot-Hawley and the Geneva round.[36]

The regression results with the 1946 AVE dependent variable are in Panel A of Table 9. The estimates on the Smoot-Hawley tariffs have the predicted positive and significant estimates that are slightly larger than the baseline results in Table 7. For each import value variable, the coefficient estimates continue to be negative and are significant in all six columns, and are larger in magnitude than the baseline results. This suggests the tariff changes prior to GATT are also consistent with the TOT theory predictions on their own.

Finding support for the TOT motive when focusing on tariff changes due to the bilateral trade agreements raises an important question: were the U.S. tariff changes from Smoot-Hawley to Geneva driven only by these bilateral trade agreements? To further explore this and as an alternative robustness check, we return to our baseline empirical approach that uses Geneva AVE tariffs as the dependent variable, but instead use the subsample of observations that did not experience any tariff changes between Smoot-Hawley and 1946.[37] The regression results are in Panel B of Table 9. The number of observations decreases to 2,907 for specifications that include import values that equal zero; the number of observations decreases further to 2,601 when using log imports. Even with these decreases in the sample

[36] $(39.45 - 26.89)/(39.45 - 20.41) = 0.6597$.

[37] We define this subsample as products with identical Smoot-Hawley and 1946 MFN tariffs (i.e., AVEs are identical when unit values are held constant at 1933 levels).

sizes, the results are similar to the baseline results with the import value estimates continuing to be negative and statistically significant. Taken together with the 1946 tariff results (Panel A), this provides strong evidence that the empirical results are driven both by bilateral negotiations and GATT negotiations.

6.3 Other Provisions Impacting Tariffs

In addition to the bilateral trade agreements, a subset of products were subject to quotas and other unilateral actions impacting U.S. tariffs prior to the GATT.[38] Appendix 10.3.2 provides descriptions of the five categories of U.S. legislative provisions that occur in the data, ranging from tariffs set under Section 336 of the Tariff Act of 1930 to those under the Sugar Act. Table 10 drops products that were subject to quotas or any of these provisions to explore if the empirical results are sensitive to these cases. The coefficient estimates for Smoot-Hawley tariffs and the import value variables are similar in magnitude and statistical significance as the baseline results in Table 7.[39]

6.4 Caps on Negotiated Tariff Changes

Under the RTAA, tariff decreases in bilateral trade agreements occurring between Smoot-Hawley and 1946—the year prior to GATT—were generally capped at 50 percent. Between 1946 and the first GATT round in 1947, U.S. trade negotiators received additional negotiating authority to reduce tariffs by up to another 50 percent from the 1946 levels. This statutory restriction had the potential to limit the magnitude of tariff liberalization over the period. Indeed, about 14 percent of products experience a tariff decrease of at least 50 percent during the pre-Geneva period, and about 19 percent of products experience a tariff decrease of at

[38] Although quotas were an important trade policy tool of major U.S. trading partners during the interwar period, the U.S. use of quotas was more limited during this period. Fewer than three dozen products have true quotas in our data.

[39] As an alternative, we use the baseline sample and include a dummy variable as a control variable that indicates if the product was subject to any quota or other provision. The coefficient estimates for Smoot-Hawley tariffs and the import value variables continue to be similar to the baseline results in Table 7. The estimates on the dummy control variable are statistically insignificant at the 10 percent level. Results are available upon request.

Table 9: Robustness Check, Impact of Pre-GATT Bilateral Trade Agreements

Panel A: Pre-GATT Tariff Changes

Dependent Variable: AVE Tariff Prior to Geneva (1946 Unit Values)	OLS			Tobit		
	(1)	(2)	(3)	(4)	(5)	(6)
Smoot-Hawley AVE Tariff	0.383***	0.364***	0.383***	0.446***	0.416***	0.444***
	(0.029)	(0.033)	(0.029)	(0.010)	(0.010)	(0.010)
1933 Import Value	-0.279***			-2.195***		
	(0.094)			(0.376)		
Log 1933 Import Value		-0.801***			-1.066***	
		(0.115)			(0.120)	
High 1933 Import Value			-3.285***			-4.188***
			(0.479)			(0.592)
Observations	4,486	4,027	4,486	4,486	4,027	4,486
R-squared	0.613	0.617	0.615			
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Panel B: Products without Tariff Changes Prior to GATT

Dependent Variable: Geneva AVE Tariff (1947 Unit Values)	OLS			Tobit		
	(1)	(2)	(3)	(4)	(5)	(6)
Smoot-Hawley AVE Tariff	0.389***	0.360***	0.388***	0.469***	0.426***	0.469***
	(0.027)	(0.027)	(0.027)	(0.011)	(0.012)	(0.011)
1933 Import Value	-0.161***			-1.796***		
	(0.043)			(0.480)		
Log 1933 Import Value		-0.560***			-0.998***	
		(0.105)			(0.138)	
High 1933 Import Value			-2.559***			-4.113***
			(0.493)			(0.694)
Observations	2,907	2,601	2,907	2,907	2,601	2,907
R-squared	0.668	0.670	0.671			
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes (both panels): Standard errors in parentheses (OLS are heteroskedasticity-robust). The log of 1933 import values drops observations that have an import value of zero. The Tobit model specifies zero as a lower limit of the dependent variable. Smoot-Hawley AVE tariffs use 1933 unit values to calculate AVEs for observations that include specific tariffs. Singleton observations are dropped. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 10: Robustness Check, Drop Products with Quotas or Other Unilateral Provisions

Dependent Variable: Geneva AVE Tariff (1947 Unit Values)	OLS			Tobit		
	(1)	(2)	(3)	(4)	(5)	(6)
Smoot-Hawley AVE Tariff	0.327***	0.300***	0.326***	0.382***	0.346***	0.382***
	(0.027)	(0.029)	(0.027)	(0.008)	(0.009)	(0.008)
1933 Import Value	-0.182***			-0.925***		
	(0.052)			(0.325)		
Log 1933 Import Value		-0.535***			-0.755***	
		(0.097)			(0.107)	
High 1933 Import Value			-2.302***			-3.071***
			(0.423)			(0.520)
Observations	4,261	3,842	4,261	4,261	3,842	4,261
R-squared	0.569	0.567	0.571			
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors in parentheses (OLS are heteroskedasticity-robust). The log of 1933 import values drops observations that have an import value of zero. The Tobit model specifies zero as a lower limit of the dependent variable. Smoot-Hawley AVE tariffs use 1933 unit values to calculate AVEs for observations that include specific tariffs. Singleton observations are dropped. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

least 50 percent during Geneva-round negotiations.[40]

It is, therefore, important to test whether our empirical results are robust to taking this institutional feature into account. We do this by considering a sample that drops products that experienced at least a 50 percent decrease in either of the two periods (Smoot-Hawley to 1946, or 1946 to the Geneva round). Table 11 shows that the regression results using this sample are similar to the baseline results in Table 7. The estimates on the Smoot-Hawley tariffs are positive and significant, and the import value coefficient estimates continue to be negative and significant.

[40] Using AVEs that use 1933 unit values, between Smoot-Hawley and 1946, 600 products had a 50 percent decrease in the tariff rate and 32 products had a tariff decrease larger in magnitude than 50 percent. These 32 products were subject to at least one provision covered in Section 6.3. Between 1946 and 1947, 871 products had a 50 percent decrease in the tariff rate and four products had a tariff decrease larger in magnitude than 50 percent. These four products were not subject to other provisions.

Table 11: Robustness Check, Drop Observations with Tariff Decreases of at Least 50 percent

Dependent Variable: Geneva AVE Tariff (1947 Unit Values)	OLS			Tobit		
	(1)	(2)	(3)	(4)	(5)	(6)
Smoot-Hawley AVE Tariff	0.320***	0.308***	0.320***	0.379***	0.358***	0.378***
	(0.027)	(0.033)	(0.027)	(0.009)	(0.010)	(0.009)
1933 Import Value	-0.218***			-1.444***		
	(0.062)			(0.340)		
Log 1933 Import Value		-0.610***			-0.918***	
		(0.102)			(0.121)	
High 1933 Import Value			-2.893***			-4.082***
			(0.468)			(0.611)
Observations	3,620	3,247	3,620	3,620	3,247	3,620
R-squared	0.601	0.608	0.604			
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors in parentheses (OLS are heteroskedasticity-robust). The log of 1933 import values drops observations that have an import value of zero. The Tobit model specifies zero as a lower limit of the dependent variable. Smoot-Hawley AVE tariffs use 1933 unit values to calculate AVEs for observations that include specific tariffs. Singleton observations are dropped. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6.5 Demand and Supply Elasticities

For the final robustness check, we use the more general specification in Equation 1 that relaxes the linear demand and supply assumptions. Compared to the baseline specification in Equation 3, we replace the import value variable (V_g) with $\eta_g \equiv (\sigma_g^{BR}/\omega_g^{*BR})(M_g^{BR}/p_g^{BR})$. The terms-of-trade theory implies a negative coefficient on η_g and a positive coefficient estimate on Smoot-Hawley tariffs.

For the best-response import demand (σ_g^{BR}) and foreign export supply (ω_g^{*BR}) elasticities, we first implement the estimation methodology from Soderbery (2015) to estimate these elasticities at the 4-digit level.[41] This approach requires import values and quantities by exporter for multiple time periods. We use the earliest available consecutive year in our data that covers U.S. imports by exporter over 1935–1938, and aggregate the 7-digit import

[41]This approach builds on Feenstra (1994) and uses an LIML regression model to estimate the elasticities. See Grant and Soderbery (2024) for further discussion on potential sources of bias related to weak instruments and violations of the exclusion restriction when estimating elasticities with an IV approach.

values and quantities to the 4-digit level for the elasticity estimation.[42] The Soderbery (2015) estimation approach yields 763 4-digit codes with feasible elasticity estimates.[43] This yields a median σ_g of 2.99 and median $\frac{1}{\omega_g^*}$ of 0.15.[44] These median values are similar to those of Irwin and Soderbery (2021), who estimate a median demand elasticity of 2.86 and median inverse foreign export supply elasticity of 0.18 using U.S. import data over 1927–1935. We then match these 4-digit elasticity estimates to all the corresponding 7-digit product codes within the respective 4-digit grouping. This implies we assume 7-digit products under each 4-digit code have the same demand and supply elasticities.

Beyond obtaining demand and supply elasticity estimates, another challenge with calculating η_g is it requires best-response import quantities relative to best-response prices. As previously noted, this is problematic as historical U.S. import data uses a wide range of units to list quantities, making comparison across product quantities challenging. We select a subsample of products that have the same—and most commonly used—unit indicator in the data—the weight in pounds. This unit classification covers over half of the 7-digit products in the sample.[45] For this subsample of products, we aggregate the 1933 product-level import values and quantities to the 4-digit level, and assume that all 7-digit products within a 4-digit grouping have the same unit value (i.e., V_g^{1933}/M_g^{1933}). We use these unit values to proxy for prices (p_g^{BR}), then calculate unit-consistent import quantities at the 7-digit level, such that $M_g^{BR} = V_g^{1933}/p_g^{BR}$.[46] This yields the required inputs (σ_g^{BR}, ω_g^{*BR}, M_g^{BR}, p_g^{BR}) to calculate η_g at the product (7-digit) level.

The regression results using η_g are in Table 12. As with our prior analysis, we consider

[42] When aggregating the trade data, we check that the unit codes across 7-digit products within a 4-digit grouping are consistent. In cases where there are differences in unit codes within a 4-digit group we use the following procedure. First, we select observations with the most commonly reported unit code within the 4-digit grouping. Second, if different unit codes within a 4-digit grouping have the same frequency, then we select the unit code that corresponds with the largest total trade value over 1935–1938.

[43] As elasticity estimation can result in some extreme outliers, we truncate the elasticity estimates. We set the maximum values for the estimate of (σ_g^{BR}) and (ω_g^{*BR}) to be 100. Also, if the estimate of (ω_g^{*BR}) is negative we set the value to 0.0001. These assumptions are similar to the approaches used by Soderbery (2015) and Grant and Soderbery (2024).

[44] The mean σ_g is 11.19 and mean $\frac{1}{\omega_g^*}$ is 1.13.

[45] While this could potentially introduce sample selection into the estimation, this approach allows for the majority of the observations to be included the analysis and records units consistently across observations.

[46] The unit values for the regression sample range from \$0.002 per pound to \$87.66 per pound, with a median value of \$0.11 per pound.

three functional forms for the η_g term: level, log, and binary.[47] The log η_g variable takes the log of η_g. The binary variable, High η_g, denotes η_g values above the median η_g value. As with the baseline model that uses import values (see Table 7) and consistent with the TOT theory predictions, the estimates on Smoot-Hawley tariffs are positive and the three functional forms of η_g are negative. These point estimates are also statistically significant at the 10 percent level for the level of η_g, and at the 1 percent level for the log and binary measures. Using a Tobit model (columns 4–6) continues to yield point estimates for the η_g terms that are larger in magnitude compared to the OLS estimates (columns 1–3), which is similar to what we found when using import values in the baseline approach. The level of η_g is insignificant, but in the preferred log and binary forms, the coefficient estimates continue to be statistically significant at the 1 percent level. The decrease in statistical significance for the level of η_g could also be due to the decrease in the sample size (from 4,643 to 2,006 observations). Overall, even given the data limitations for obtaining η_g, the estimation results are supportive of the TOT predictions.

7 Conclusion

This paper provides the first empirical test of the terms-of-trade theory for a large country moving from unilateral tariffs to cooperative trade agreements. We compile and concord a new product-level data set on U.S. tariffs and imports spanning the Smoot-Hawley Tariffs of 1930 through the first GATT round in 1947. Given the prevalence of specific tariff rates over this period, we calculate ad valorem equivalents that allow for straightforward comparisons across different types of tariffs. Making this data available is a contribution in its own right, supporting future academic research and policy analysis on this pivotal episode in U.S. trade history.

Using this data set, we document several stylized facts about U.S. trade policy in this period. Tariffs decreased substantially across products and industry groups, with bilateral

[47]We divide the level of η_g by 1 trillion to help with the presentation of the small regression coefficient estimate. The summary statistics for these three variables are in Table 6. The number of observations in Table 12 slightly decreases compared to the number of observations listed in Table 6, as we drop singleton observations from the regressions.

Table 12: Robustness Check, Alternative Terms-of-Trade Specification

Dependent Variable: Geneva AVE Tariff (1947 Unit Values)	OLS			Tobit		
	(1)	(2)	(3)	(4)	(5)	(6)
Smoot-Hawley AVE Tariff	0.226***	0.200***	0.220***	0.278***	0.245***	0.270***
	(0.025)	(0.023)	(0.025)	(0.012)	(0.012)	(0.012)
η_g	-0.213*			-2.921		
	(0.125)			(2.192)		
Log η_g		-0.337***			-0.525***	
		(0.056)			(0.061)	
High η_g			-3.451***			-5.128***
			(0.625)			(0.727)
Observations	2,006	1,884	2,006	2,006	1,884	2,006
R-squared	0.527	0.539	0.534			
Industry Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors in parentheses (OLS are heteroskedasticity-robust). The log of η_g drops observations that have a value of zero. The Tobit model specifies zero as a lower limit of the dependent variable. Smoot-Hawley AVE tariffs use 1933 unit values to calculate AVEs for observations that include specific tariffs. Singleton observations are dropped. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

agreements flowing from the Reciprocal Trade Agreements Act of 1934 accounting for roughly half of the total liberalization achieved by 1947. Despite inflation eroding the protective value of specific tariffs set in 1930, U.S. negotiators did not systematically convert specific tariffs into ad valorem ones—the shift toward ad valorem rates that characterizes modern U.S. tariffs came later.

We then extend the empirical approach of Bagwell and Staiger (2011) to test whether U.S. tariff changes were consistent with predictions from the terms-of-trade theory. The estimation results provide strong support: conditional on Smoot-Hawley tariffs and industry fixed effects, GATT tariffs are lower when the U.S. imports more of the product. The estimated effect is larger than that found in studies of developing-economy WTO accession, consistent with the U.S. having greater importer market power. This support for the theory holds despite institutional constraints—the RTAA caps, the principal supplier rule, and the gradual nature of the liberalization—that should have made the terms-of-trade results more difficult to detect.

Avenues for future research include exploring the TOT theory in the context of other developed economies, further studying both the motivations and impacts of U.S. trade liberalization in this era, and comparing U.S. behavior in this period with that of other large economies entering cooperative trade regimes. As policymakers reconsider broad-based protection in the present moment, understanding how the United States dismantled the previous peak in tariff protection—and why it did so in the way it did—has acquired new practical relevance.

8 Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used claude.ai in order to assist with writing code to clean the dataset and refine some figures and tables. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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10 Data Appendix

10.1 Data sources

Data sources information is listed in Table 13. In the below subsection, we briefly explain how we collect and merge multiple data sources to construct the final data set for analysis. We provide further information regarding the details of digitization and cleaning of these historical documents in the Supplementary Appendix for readers that are interested.

10.1.1 Tariff Commitment Data

The United Nations Treaty Collection (UNTC) website houses the detailed GATT tariff schedules of every country of each round of GATT negotiations from the first Geneva Round to the Uruguay Round.[48] We take this to be the definitive version of the U.S. tariff schedule for Geneva 1947.

To the Geneva 1947 tariff data, we add the tariff rates from the original text of the Smoot-Hawley Act as well as those from *United States Import Duties (June 1946)*, which is a list of all the rates of duty as of just before the Geneva round of the GATT. The 1946 tariffs were compiled by the U.S. Tariff Commission, the forerunner to the USITC, to give U.S. negotiators the correct baseline from which to make concessions in the first round of the GATT. These documents are all organized using the paragraph numbers in the Tariff Act of 1930, which has 966 paragraphs, where many paragraphs cover multiple items with different tariff rates.[49] The number of distinct tariff rates within paragraphs grows over time as tariff commitments are often made so that items that previously shared a tariff see their tariff rates diverge. As new negotiations are concluded, products previously listed as one item—that had a single tariff rate—are often split into multiple items with different tariff rates.[50] We use the term 'item' or 'product' to refer to an observation in our data set.

We ultimately reorganize this tariff data according to the United States' Schedule A

[48] Digitizing additional GATT rounds is an area for future work. Given this paper's focus on the U.S. moving from a non-cooperative to cooperative trade policy setting between Smoot-Hawley and the beginning of the GATT, we focus on the Geneva 1947 round.

[49] These paragraph numbers were established in the Tariff Act of 1922.

[50] See our discussion in supplementary appendix and Acosta and Cox (2024) for in-depth examples.

classification system, which was used to classify import data until 1963. We did this for two reasons. First, although this import classification system is not perfectly hierarchical, the organization of Schedule A is far superior to that of the Smoot-Hawley legislation for creating a data set on which one can reliably perform statistical analysis. Second, we want to integrate import data into the tariff data so that we can calculate AVEs.

While the Schedule A evolves over time, we have organized the tariff data using the Schedule A numbers when the Geneva Round went into effect in 1948.[51] We use the cross-reference in the 1946 Schedule A, along with the Smoot-Hawley and Schedule A descriptions, to manually reorganize our tariff commitment data set by Schedule A number.

10.1.2 Import Data

We also require U.S. import data to create AVEs for descriptive statistics and the TOT analysis. We have acquired complete import data—including exporter detail—for 1935–1938 and 1947.[52] We also manually gathered total import quantities and values for 1933 and 1946.

The data source for imports by Schedule A change over time. We acquired PDF versions of the Department of Commerce publication *Foreign Commerce and Navigation of the United States* for each calendar year from 1929 to 1946. These documents contain not only import quantity, unit of measure, and value, but also the tariff rate, duties collected, and AVEs for specific tariffs in some years. This document was discontinued after 1946, so for 1947, we use PDF versions of the Census publication *United States Imports for Consumption of Merchandise, Commodity by Country of Origin (FT 110)* that provide tables of import values, quantities, and units of measure.

We use the Consumption of Merchandise series throughout for two reasons. First, they are available for every year until 1962, unlike the General Imports of Merchandise tables,

[51] We determine this using a combination of the 1946 and 1950 Schedule A documents, which include "Commodity numbers", which we refer to as Schedule A numbers; product descriptions, units of quantity, Smoot-Hawley tariff levels, Smoot-Hawley paragraph numbers, and the MFN tariffs levels. They also include the source of each tariff change between 1930 and 1946, be it unilateral or through a bilateral trade agreement that became an MFN tariff.

[52] All of these except the 1935 data were previously compiled under NSF Project 1326940 and graciously shared with us by Bob Staiger and Tricia Mueller. We contracted with an external service to help us digitize the 1935 data.

which we could not find for all years. Second, the Consumption of Merchandise series represents the imports on which tariffs are applied. The primary difference between the two series is that the General Imports of Merchandise tables include imports that entered bonded warehouses in the U.S. but did not enter the U.S.'s customs territory during the same year.[53]

10.2 Calculating AVEs

To analyze all tariffs on the same basis, we calculate the AVEs for all products with a specific-tariff component by combining the tariff commitment data with the import data.[54] We first calculate unit values by dividing import values by import quantities, ensuring that the specific tariff units and import quantity units align for each product.[55] For compound tariffs, we then add the ad valorem component to the AVE of the specific component. The process is a bit more complex for products with 'mixed' tariffs, where the tariff is either specific or ad valorem, with the given rate bracketed by minimum and/or maximum rates of the other type. Therefore, we calculate the AVE of the specific component(s) and compare to the ad valorem component(s) to determine which rate applied. We provide more details in the supplementary appendix on mixed tariffs.

10.3 Notable Data Details

10.3.1 Free List

Many of the duty-free products in Schedule 16 of Smoot-Hawley remained unconstrained by trade commitments for many years. While the U.S. unilaterally set the Smoot-Hawley tariffs and could have increased them at any time, the only cases we found of a Free-list product receiving a strictly positive tariff up to 1947 are for a handful of sugar products whose tariffs were increased under the Sugar Act (see Section 10.3.2 below for details). In

[53]Irwin and Soderbery (2021) use the General Imports of Merchandise series for the late 1920s and early 1930s, where full country-product detail is available.

[54]For Smoot-Hawley AVEs, we use 1933 unit values; for 1946 AVEs, we use 1946 unit values; for Geneva AVEs, we use 1947 unit values. In cases where we are unable to calculate unit values (e.g., if import values or quantities are missing or are zero), we do not attempt to calculate an AVE. Around 700 products are impacted by this issue.

[55]See Appendix C of Teti (2020) for a clear and concise explanation of conversion of specific tariffs into AVEs.

Table 13: Summary of Data Sources

Document	Content	Years	Sources
Consolidated Schedule of Tariff Concessions	Consolidated Schedule of Geneva, Annecy and Torquay	1952	United Nations Publications. International Documents Service, Columbia University Press
United Nations Treaty Collections (UNTC)	Individual negotiated agreements in GATT in terms of the Smoot-Hawley Paragraph Number	1947 (Geneva), 1948 (Annecy), 1950 (Torquay)	UNTC official website
The United States Senate Library series	The Smoot-Hawley Act in 1930	1930	Federal Register
Statistical Classification of Imports into the United States	Executive tariff rates implemented by the U.S. government in terms of the Schedule A code.	1930, 1933, 1936, 1937, 1939, 1941, 1943, 1946, 1950	U.S. Department of Commerce, Bureau of the Census
Agreement between the United States and [country] respecting reciprocal trade	Full text of each bilateral trade agreement, including schedules of concessions	1935–1939, 1942-1944	Federal Register
The Foreign Commerce and Navigation of the United States	Imports of products into the U.S. documented in terms of the Schedule A code	1933, 1935, 1946	U.S. Department of Commerce, Bureau of the Census
United States Imports for Consumption of Merchandise	Imports of products into the U.S. documented in terms of the Schedule A code	1947	U.S. Department of Commerce, Bureau of the Census

contrast, we observe hundreds of cases where Smoot-Hawley duty-free rates became bound during bilateral trade agreements or the GATT. We denote Smoot-Hawley Free-list products with a “0” for the ad valorem tariff.

Under the Smoot-Hawley paragraph number classification system, duty-free items were gathered into Schedule 16 instead of integrated into a schedule with similar products. We have integrated these duty-free items into our data and organized by Schedule A so that all like products appear together.

10.3.2 Unilateral Actions Impacting U.S. Tariffs

The provisions listed below cover five classes of tariff changes that were primarily made by unilateral U.S. policy. We detect which products were subject to these provisions in the Schedule A column "1930 Tariff Act (except as noted)," where the notes indicate any provision which makes the statutory tariff different from the Smoot-Hawley level. Products that are subject to these provisions have non-cooperative rates (the current 'Column 2' rates) different from the original Smoot-Hawley rates.

Section 336: The flexible tariff provision (Section 336 of the Tariff Act of 1930) gave the President of the United States authority to unilaterally change individual import duties, that is, to increase or decrease tariffs by up to 50 percent. Our data shows 183 products impacted by Section 336 changes.

Sugar Act: The goal of the Sugar Act was to control sugar imports into the U.S. and stabilize the sugar sector. The Sugar Act offered short-term solutions to control the sugar market and preserve price stability, and included establishing quotas and tariffs on sugar. We see several revisions to the Sugar Act during our time period, with 80 products affected.

Quotas: Thirty-six products have the notation "quota" in the Schedule A as of 1946. Some are tariff-rate quotas, so an even smaller number of products have true quotas.

Internal Revenue Code (IRC): The IRC serves as the comprehensive body of law that consolidates all federal tax statutes, covering a wide range of taxes such as income tax, estate tax, gift tax, excise tax, alcohol tax, tobacco tax, and employment tax. Before the IRC's existence, U.S. tax laws were published in separate volumes based on subject matter. The Internal Revenue Code went into effect in 1939, but internal tax provisions were applied to imports in the years before the code was consolidated. We refer to all such taxes as IRC provisions. IRC rates were added to the tariff for all products with a domestic excise tax to ensure foreign producers did not escape the excise tax and gain an advantage over domestic producers. IRC rates were sometimes included in bilateral negotiations. There are

309 products in our data that were impacted by internal tax provisions.

Emergencies: Section 318 of the Tariff Act of 1930 granted authority to the President, the Secretary of the Treasury, and the Commissioner of Customs to change tariffs during a state of emergency. When the President declares an emergency due to war or other reasons, they can authorize the Secretary of the Treasury to extend the time allotted for certain activities and allow duty-free importation of essential supplies for emergency relief work. The Secretary of the Treasury was required to report to Congress about any actions taken under this provision.[56] There are 5 products in our data that had emergency tariffs at some point.

The American Selling Price: This provision allowed the tariff to be calculated based on domestic prices instead of the price charged by foreign exporters on some products. It was used primarily for chemical products. The American Selling Price was applied to 43 products in our data set.

[56] See https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title19-section1318&num=0&edition=prelim.